

Question

Write a Java program to implement bank functionality where RBI controls minimum interest rate, minimum balance, and withdrawal limits. Demonstrate dynamic polymorphism using classes Customer, Account, RBI (Base Class), and derived classes (SBI, ICICI, PNB).

Steps for Execution

1. Open your Java IDE (e.g., Eclipse) or a text editor.
2. Create a new file named BankDemo.java.
3. Copy the program code into the file.
4. Save the file.
5. Compile the program using:

```
javac BankDemo.java
```
6. Run the program using:

```
java BankDemo
```
7. Enter the bank name when prompted and observe the output.

Algorithm

1. Define a base class RBI with method `getRateOfInterest()` returning 4%.
2. Create derived classes SBI, ICICI, and PNB overriding `getRateOfInterest()` with their own values.
3. Create a Customer class to hold customer details.
4. In BankDemo main class:
 - Accept bank name input from user.
 - Use a base class reference (RBI bank) to point to the chosen derived class object.
 - Call `getRateOfInterest()` method.
5. Display the interest rate based on the selected bank.

UML Class Diagram

Code:

```
import java.util.Scanner;
```

```
class RBI {  
    public double getRateOfInterest() {  
        return 4.0;  
    }  
}
```

```
class SBI extends RBI {  
    @Override  
    public double getRateOfInterest() {  
        return 7.0;  
    }  
}
```

```
class ICICI extends RBI {  
    @Override  
    public double getRateOfInterest() {  
        return 6.5;  
    }  
}
```

```
class PNB extends RBI {  
    @Override  
    public double getRateOfInterest() {  
        return 6.0;  
    }  
}
```

```
class Customer {  
    String name;  
    String bankName;  
    Customer(String name, String bankName) {  
        this.name = name;  
        this.bankName = bankName;  
    }  
}
```

```
public class BankDemo {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter the Bank name to find the rate of Interest : ");  
        String bankName = sc.nextLine();  
  
        RBI bank;  
        switch (bankName.toUpperCase()) {  
            case "SBI": bank = new SBI(); break;  
            case "ICICI": bank = new ICICI(); break;  
            case "PNB": bank = new PNB(); break;  
            default: bank = new RBI(); break;  
        }  
  
        System.out.println(bankName + " rate of interest is : " + bank.getRateOfInterest() +  
"%");  
        sc.close();  
    }  
}
```

}

Output:

Enter the Bank name to find the rate of Interest : RBI

RBI rate of interest is : 4%

Enter the Bank name to find the rate of Interest : RBI

RBI rate of interest is : 4%